Scale-Aware Mamba-Driven Implicit Representation for Arbitrary-Resolution Hyperspectral Pan-Sharpening

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***Abstract*—Pan-sharpening aims to reconstruct high-resolution hyperspectral images (HR-HSI) by fusing the spectral information of hyperspectral images (HSI) with the spatial details provided by panchromatic (PAN) images. However, most existing pan-sharpening methods regard the fusion of hyperspectral and PAN images at different spatial resolutions as independent tasks, necessitating the training of a distinct model for each scale. This paradigm is impractical in real-world applications, as maintaining separate models for every spatial scale incurs prohibitive computational and storage costs in arbitrary-resolution pan-sharpening scenarios. Therefore, we propose the scale-aware Mamba-driven implicit representation network (SAMIR-Net), a unified scale-aware framework for arbitrary-resolution hyperspectral pan-sharpening. The proposed SAMIR-Net consists of two main components: scale-aware modality interactive extraction and implicit spatial–spectral reconstruction. The scale-aware modality interactive extraction is realized through a dual-branch, scale-sensitive subnetwork that collaboratively extracts scale-aware multimodal features and facilitates spatial–spectral interaction. The implicit spatial–spectral reconstruction module (IS^{2}RM) further performs cross-attention between spatial and spectral embeddings, enabling flexible and robust coordinate-based pan-sharpening at arbitrary resolutions. Extensive experiments show that SAMIR-Net achieves favorable performance compared with representative state-of-the-art methods and exhibits improved generalization across different upsampling factors, indicating its potential for flexible hyperspectral reconstruction. The code is available at: https://github.com/Jiahuiqu/SAMIR-Net**

***Index Terms*—Arbitrary-resolution reconstruction, hyperspectral pan-sharpening, implicit neural representation (INR), Mamba, scale-aware learning, spatial–spectral fusion.**

I. Introduction

HIGH-RESOLUTION hyperspectral images (HR-HSI) provide rich spatial–spectral information and are highly valuable for remote sensing applications. However, their acquisition remains limited by the tradeoffs among spatial resolution, spectral resolution, and signal-to-noise ratio in optical imaging systems. Compared with single-modality reconstruction methods [1], [2], [3], multimodal approaches can better exploit complementary information from heterogeneous sensors. Related multimodal remote sensing studies have demonstrated the effectiveness of cross-modal interaction for representation enhancement [4], [5], [6], [7], [8], [9], [10], [11]. Among these paradigms, hyperspectral pan-sharpening has emerged as a practical solution for reconstructing HR-HSI by fusing the spectral information of hyperspectral images (HSI) with the high-resolution spatial details of panchromatic (PAN) images. These advances continue to support a variety of remote sensing applications, including classification, environmental monitoring, and change detection [12], [13], [14], [15], [16], [17], [18], [19], [20], [21].

Recent surveys and reviews [22], [23] indicate that hyperspectral pan-sharpening has progressed from classical image fusion models to deep representation learning. Early methods mainly rely on hand-crafted priors and fixed assumptions [24], [25]. CS methods, such as principal component analysis [26] and Gram–Schmidt [27], are efficient but prone to spectral artifacts. MRA-based methods [28], [29] enhance spatial details through frequency decomposition but may weaken spectral consistency. Bayesian approaches [30] depend heavily on observation models and priors, while matrix factorization methods [31], [32] are limited in modeling complex cross-modal and scale-dependent relationships. Overall, these model-driven methods often suffer from spectral distortion, noise, and hyperparameter sensitivity, and limited capacity for cross-modal interaction and scale-varying nonlinear modeling, which restricts their robustness and generalization.

Deep learning-based pan-sharpening has achieved substantial progress by learning nonlinear mappings from data. Early CNN-based methods, such as PNN [33], demonstrated the effectiveness of convolutional networks for spatial detail injection. Subsequent studies further improved hyperspectral reconstruction through tailored designs, including improved deep image prior with residual reconstruction [34], local-context adaptive convolution with global harmonic bias [35], and diffusion-driven matching and fusion for reference-based HSI super-resolution [36]. Despite these advances, convolution-based methods still have limited capacity to

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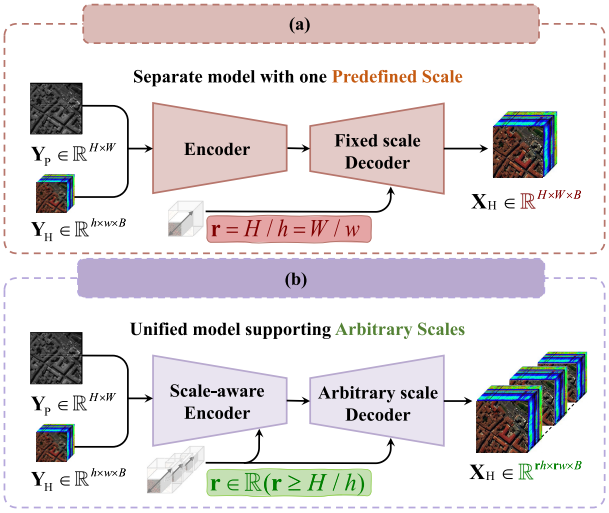


Fig. 1. Comparison of (a) traditional pan-sharpening fusion models, which are limited to fixed upsampling ratios and (b) the proposed arbitrary-resolution fusion model. Unlike conventional approaches, our method is capable of pan-sharpening HSIs to any desired spatial scale with one single model, enabling flexible and high-fidelity reconstruction at arbitrary resolution.

model long-range dependencies and complex cross-modal interactions.

Selective state-space models (SSMs) [37], represented by Mamba [38], provide an efficient alternative to attention with linear complexity and strong global modeling. Recent visual extensions further support hierarchical spatial modeling across scales [39], [40], [41]. FusionMamba [42] and Pan-Mamba [43] have shown the potential of SSMs for spatial–spectral integration in image fusion. These properties make SSMs suitable for pan-sharpening tasks that require efficient and robust cross-modal modeling.

In practical scenarios, the desired output resolution is often task-dependent and may exceed the PAN resolution. Such reconstruction can support fine-scale land-cover analysis, object-level interpretation, change monitoring, and boundary delineation by providing denser hyperspectral spatial sampling [44], [45], [46]. Nevertheless, most existing methods train separate models for different spatial scales, as shown in Fig. 1, leading to high computational and storage costs when multiple target scales are required. Arbitrary-resolution hyperspectral pan-sharpening provides a scale-flexible alternative by enabling one model to satisfy diverse resolution requirements [47]. Related arbitrary-scale super-resolution studies, including Meta-SR [48], CNN-based relay optimization [45], and variable subpixel convolution [46], have demonstrated the feasibility of scale-controllable reconstruction. In addition, implicit neural representations (INRs) support arbitrary-resolution prediction by mapping continuous coordinates to signal values, and have been widely studied in natural image reconstruction [49], [50], [51], [52], [53]. However, INR-based pan-sharpening remains underexplored, with only a few recent attempts [54], [55].

Despite these advances, arbitrary-resolution hyperspectral pan-sharpening is not merely a decoder-side coordinate interpolation problem. Existing methods usually treat the target scale as an output-side variable, while insufficiently adapting feature extraction to scale variation. This is important because PAN-derived spatial structures and HSI spectral signatures respond differently to changing resolutions, and complex regions such as urban boundaries are more sensitive to scale-induced distortion than homogeneous areas. Therefore, coordinate-wise decoding alone may be insufficient for learning scale-sensitive multimodal representations.

To address these limitations, we propose the scale-aware Mamba-driven implicit representation network (SAMIR-Net), a unified framework for arbitrary-resolution hyperspectral pan-sharpening. Different from prior methods that mainly emphasize continuous output generation, SAMIR-Net explicitly incorporates the target scale into feature extraction, cross-modal interaction, and implicit reconstruction. Specifically, SAMIR-Net consists of two main components: scale-aware modality interactive extraction and implicit spatial–spectral reconstruction. The former embeds scale factor priors through the scale-aware Mamba extraction block (SMEB) and promotes spatial–spectral interaction through the spatial–spectral Mamba interaction block (S^2MIB), while the latter performs coordinate-based prediction for flexible reconstruction at arbitrary resolutions.

The main contributions are summarized as follows.

1) We formulate arbitrary-resolution hyperspectral pan-sharpening as a scale-conditioned multimodal representation learning problem, rather than only a coordinate-querying problem, and establish a unified scale-aware Mamba-driven architecture that couples scale-aware modality interactive extraction with implicit spatial–spectral reconstruction.
2) We develop a scale-aware modality interactive extraction module that explicitly embeds the target scale into feature extraction and adaptively models cross-modal interactions, enabling the network to capture scale-sensitive structures before implicit reconstruction.
3) We propose an implicit spatial–spectral representation module, which exploits the potential of INR for coordinate-conditioned prediction in both spatial and spectral domains, enabling flexible, pixel-wise hyperspectral reconstruction at any target scale.

II. Related Work

A. Hyperspectral Pan-Sharpening

Early advances in hyperspectral pan-sharpening build on multispectral fusion methodologies and adapt deep models to the high spectral dimensionality of HSI. HyperPNN [56] introduced a hyperspectral-specific framework with spectrally predictive CNNs to improve spectral fidelity. Along this line, DHP-DARN [57] integrates a deep hyperspectral prior with a dual-attention residual network, and Bandara et al. [34] improved deep-prior-based hyperspectral pan-sharpening through residual reconstruction. Similarly, the enhanced deep image prior framework in [7] further shows the effectiveness of deep priors for hyperspectral spatial detail recovery. Beyond prior-driven designs, architecture-level mechanisms have been developed to better organize spectral–spatial fusion. U2Net [58] provides a spatial–spectral-integrated double U-

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Net framework, LPPNet [59] embeds deep networks into a Laplacian pyramid for hierarchical decomposition, and HyperTransformer [60] leverages Transformer interactions to enhance spatial and spectral consistency. More recently, state-space-model-based methods have improved efficiency and long-range dependency modeling in remote sensing fusion, with FusionMamba [42] proposing a general framework and pan-Mamba [43] tailoring Mamba-style cross-modal interaction to pan-sharpening. While many of these methods focus on fixed-scale fusion, arbitrary-resolution pan-sharpening enables flexible enhancement at desired scales. Although substantial progress has been made in arbitrary-scale hyperspectral pan-sharpening [45], [54], [55], [61], [62], most existing methods do not explicitly incorporate scale awareness during feature extraction, which limits their ability to capture scale-sensitive features in complex scenes.

B. Implicit Neural Representation

INR provide a continuous signal modeling paradigm that maps spatial or spatiotemporal coordinates directly to signal values through a neural function. Compared with discrete grid-based representations, INR better aligns with the continuous nature of real-world signals and allows resolution-free sampling, where the level of detail can be controlled through coordinate querying and function parameterization. This paradigm was initially popularized in 3-D scene modeling by NeRF [63], which learns a neural radiance field that maps 3-D locations and viewing directions to radiance and density for novel-view synthesis. More recently, INR has been extended to 2-D vision tasks. LIIF [49] learns a local implicit image function that predicts RGB values at continuous coordinates conditioned on nearby deep features, enabling arbitrary-resolution image reconstruction. LTE [50] further improves high-frequency recovery by introducing a local texture estimator that provides dominant-frequency cues, mitigating the difficulty of standard MLP parameterizations in representing fine textures. LIT [52] incorporates attention mechanisms and frequency encoding into a local implicit function, and uses cross-scale local attention to aggregate context across scales for arbitrary-scale super-resolution. HIIF [53] advances continuous super-resolution with hierarchical encoding and an implicit attention design to better capture fine details across multiple scales while incorporating additional nonlocal information. Despite these successes on natural images, the application of INR to HSI pan-sharpening remains limited, and training separate models for each scale is impractical. Therefore, developing a single INR-based model that generalizes across scales for multimodal fusion and high-frequency detail recovery remains an important and promising research direction.

III. Proposed Method

A. Overall Framework

Given a low-resolution HSI (LR-HSI) $\mathbf{Y}_{\mathrm{H}} \in \mathbb{R}^{h\times w\times B}$ and a high-resolution PAN image $\mathbf{Y}_{\mathrm{P}} \in \mathbb{R}^{H\times W}$, arbitrary-resolution hyperspectral pan-sharpening aims to reconstruct $\hat{\mathbf{X}}_{\mathrm{H}} \in \mathbb{R}^{rh\times rw\times B}$ at a user-specified spatial resolution, where B is the number of spectral bands and r denotes the overall upsampling factor relative to the LR-HSI. We define the fixed PAN-to-LR-HSI resolution ratio as $r_0 = H/h = W/w$, and denote by r' the target scale relative to the PAN image. The overall output scale is therefore given by $r = r_0 r'$. Under this formulation, conventional fixed-scale pan-sharpening corresponds to a fixed r', whereas arbitrary-resolution reconstruction can be achieved by varying r' without retraining the model. For $r' = 1$, the task reduces to conventional PAN-resolution pan-sharpening. For $r' > 1$, the output exceeds the PAN sampling resolution and should be interpreted as PAN-guided hyperspectral spatial super-resolution, where sub-PAN details are inferred from PAN structures and learned spatial–spectral priors rather than directly observed.

As illustrated in Fig. 2, SAMIR-Net follows a two-stage extraction-and-reconstruction pipeline. The scale-aware modality interactive extraction module first takes the interpolated LR-HSI $\widetilde{\mathbf{Y}}_{\mathrm{H}}$, the PAN image $\mathbf{Y}_{\mathrm{P}}$, and the target scale factor r as inputs to extract scale-aware HSI and PAN feature maps. The implicit spatial–spectral reconstruction module (IS^{2}RM) then performs query-wise prediction on the target output grid by using the extracted features, local relative coordinates, and pixel cell size. Specifically, the scale-aware modality interactive extraction module produces the HSI and PAN feature maps as

$$(\mathbf{F}_{\mathrm{H}}, \mathbf{F}_{\mathrm{P}}) = \Psi\left(\widetilde{\mathbf{Y}}_{\mathrm{H}}, \mathbf{Y}_{\mathrm{P}}, r; \theta_E\right) \tag{1}$$

where $\Psi(\cdot)$ is parameterized by θ_E. For each queried coordinate $\mathbf{c}_q$ on the target output coordinate grid $\mathcal{H}$, the IS^{2}RM predicts a residual spectral vector

$$\Delta\hat{\mathbf{y}}_q = \Phi\left(\mathbf{F}_{\mathrm{H}}, \mathbf{F}_{\mathrm{P}}, \delta\mathbf{c}_q, \mathbf{s}; \theta_R\right) \tag{2}$$

where $\Phi(\cdot)$ is parameterized by θ_R, $\Delta\hat{\mathbf{y}}_q \in \mathbb{R}^B$ denotes the residual correction to the interpolated LR-HSI value at $\mathbf{c}_q$, $\delta\mathbf{c}_q$ denotes the relative coordinates between $\mathbf{c}_q$ and its local supporting coordinates, and $\mathbf{s}$ is the pixel cell size of the target output grid. The final spectral prediction is obtained by

$$\hat{\mathbf{y}}_q^{\text{final}} = \widetilde{\mathbf{Y}}_{\mathrm{H}}\left(\mathbf{c}_q\right) + \Delta\hat{\mathbf{y}}_q. \tag{3}$$

The final reconstructed HR-HSI is obtained by assembling all final spectral predictions

$$\hat{\mathbf{X}}_{\mathrm{H}} = \left\{\hat{\mathbf{y}}_q^{\text{final}} \,\middle|\, \mathbf{c}_q \in \mathcal{H}\right\}. \tag{4}$$

For clarity, Algorithm 1 summarizes the complete forward procedure of SAMIR-Net, including scale computation, feature extraction, query-wise reconstruction, and HR-HSI assembly. This formulation enables SAMIR-Net to jointly exploit spatial, spectral, and scale priors, thereby supporting robust hyperspectral pan-sharpening over a continuous range of output resolutions.

B. Scale-Aware Modality Interactive Extraction

As illustrated in Fig. 2, $\Psi(\widetilde{\mathbf{Y}}_{\mathrm{H}}, \mathbf{Y}_{\mathrm{P}}, r; \theta_E)$ denotes the scale-aware modality interactive extraction module. By stacking scale-aware Mamba extraction blocks and S^{2}MIBs, it extracts scale-aware multimodal features and aligns PAN spatial details with LR-HSI spectral structures for arbitrary-resolution hyperspectral pan-sharpening.

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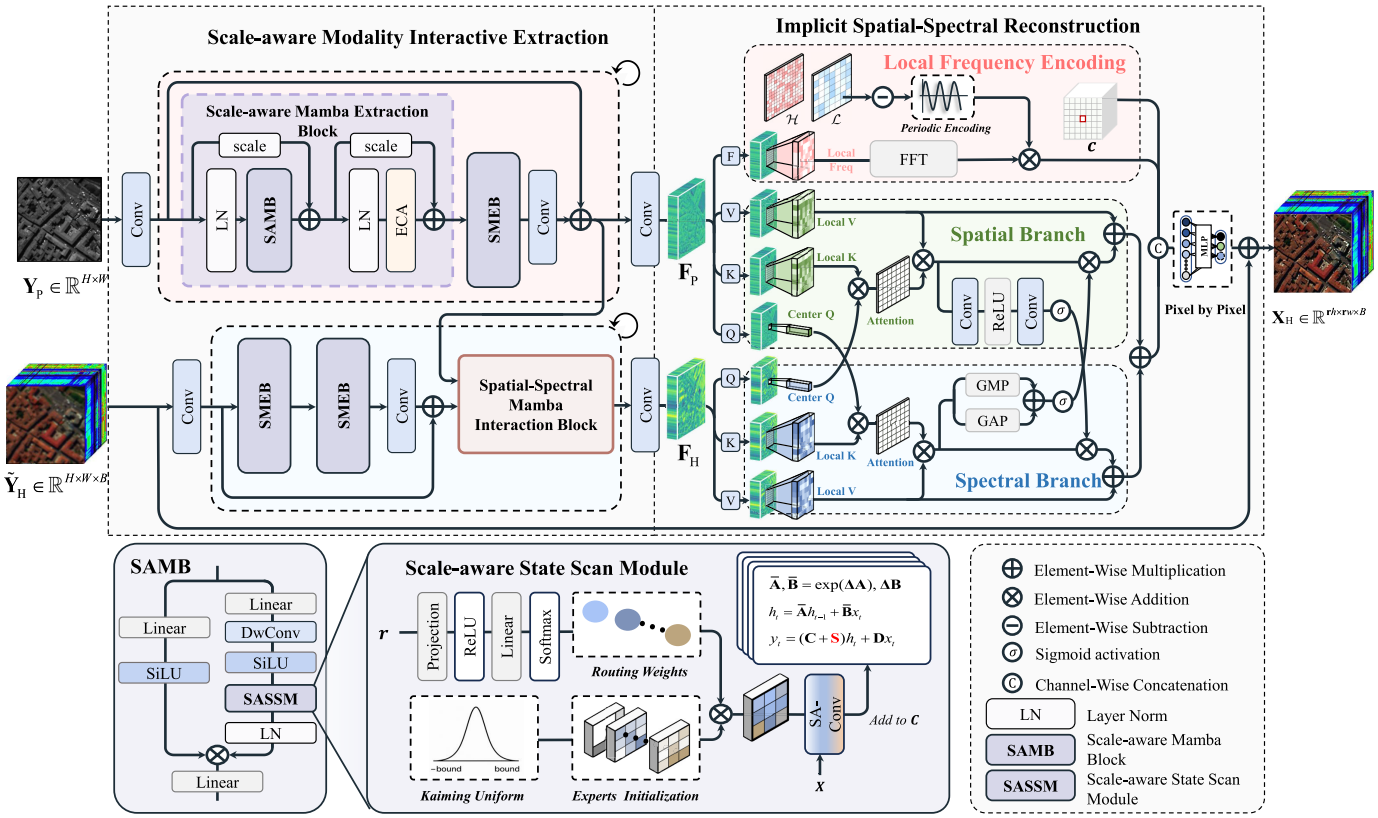


Fig. 2. Overall architecture of SAMIR-Net. The proposed framework integrates explicit scale encoding, dual-branch cross-modality feature extraction, and implicit spatial–spectral reconstruction to achieve robust and adaptive performance across diverse scaling scenarios. The bottom subfigures detail the scale-aware modules.

Algorithm 1 Forward Procedure of SAMIR-Net

Require: LR-HSI $\mathbf{Y}_{\mathrm{H}}$, PAN image $\mathbf{Y}_{\mathrm{P}}$, fixed ratio r_0, target scale r'.

Ensure: Reconstructed HR-HSI $\hat{\mathbf{X}}_{\mathrm{H}}$.

Compute $r = r_0 r'$ and construct the target grid $\mathcal{H}$ with cell size $\mathbf{s}$.

Interpolate $\mathbf{Y}_{\mathrm{H}}$ as $\widetilde{\mathbf{Y}}_{\mathrm{H}}$.

Extract scale-aware features:

$$(\mathbf{F}_{\mathrm{H}}, \mathbf{F}_{\mathrm{P}}) = \Psi(\widetilde{\mathbf{Y}}_{\mathrm{H}}, \mathbf{Y}_{\mathrm{P}}, r; \theta_E).$$

for each $\mathbf{c}_q \in \mathcal{H}$ **do**

  Compute the local relative coordinates $\delta\mathbf{c}_q$.

  Predict the residual spectral vector:

$$\Delta\hat{\mathbf{y}}_q = \Phi(\mathbf{F}_{\mathrm{H}}, \mathbf{F}_{\mathrm{P}}, \delta\mathbf{c}_q, \mathbf{s}; \theta_R).$$

  Obtain the final spectral prediction:

$$\hat{\mathbf{y}}_q^{\text{final}} = \widetilde{\mathbf{Y}}_{\mathrm{H}}(\mathbf{c}_q) + \Delta\hat{\mathbf{y}}_q.$$

end for

Assemble $\{\hat{\mathbf{y}}_q^{\text{final}} \mid \mathbf{c}_q \in \mathcal{H}\}$ as $\hat{\mathbf{X}}_{\mathrm{H}}$.

return: $\hat{\mathbf{X}}_{\mathrm{H}}$.

1) Scale-Aware Mamba Extraction Block: The effect of scale variation on reconstruction quality is region-dependent. Complex regions, such as urban areas, are more sensitive to distortion, whereas homogeneous regions, such as water bodies, are relatively stable. This motivates the scale-sensitivity analysis in Fig. 3. At the image level, two representative patches are evaluated across multiple scale factors. For each metric, including PSNR, CC, SAM, RMSE, Q2n, and ERGAS, the absolute normalized variation relative to the initial scale is integrated along the scale axis to obtain the area under the curve (AUC), where a larger AUC indicates higher scale sensitivity. At the pixel level, the RMSE variation across scales is similarly integrated to generate pixel-wise AUC heatmaps. As shown in Fig. 3, the urban patch shows strong sensitivity around structural boundaries, while the water patch exhibits sparse responses. These results indicate that scale-induced degradation is structure-dependent,

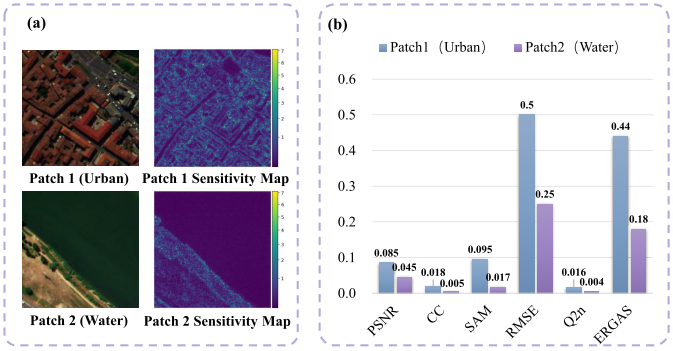


Fig. 3. Region-dependent scale sensitivity. (a) Pixel-wise AUC heatmaps from per-pixel RMSE across scales. (b) Image-level AUC of normalized metric changes.

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highlighting the need for scale-aware feature extraction in arbitrary-resolution hyperspectral pan-sharpening.

To address this issue, we propose the SMEB, which captures global context and local scale-sensitive structures from hyperspectral and PAN inputs. As illustrated in Fig. 2, each SMEB consists of a normalization layer, a scale-aware Mamba block (SAMB), and an efficient channel attention module. Learnable residual scaling factors are used to fuse transformed and input features, promoting stable feature propagation across scales.

Within SAMB, the scale-aware state space module (SASSM) serves as the core operator for noncausal and scale-sensitive sequence modeling. To enhance structure-sensitive responses under different scale factors, we introduce an explicit scale-aware modulation term into the output transformation. Given the intermediate feature map $\mathbf{F}_{\mathrm{M}}$ of modality M, we first flatten its spatial dimensions into a feature sequence $\mathbf{x} \in \mathbb{R}^{L\times C}$, where L denotes the flattened sequence length and C is the channel dimension. A scale-aware enhancement term is computed as

$$\mathbf{S} = \mathrm{Conv}_{\mathrm{scale}}(\mathbf{x}) \tag{5}$$

where $\mathbf{S} \in \mathbb{R}^{L\times d}$, d denotes the hidden-state dimension of the state-space model, and $\mathrm{Conv}_{\mathrm{scale}}(\cdot)$ denotes a scale-conditioned convolution operator. Its kernel is formed by a weighted combination of scale-dependent expert kernels, with routing weights dynamically determined by the target scale factor r and shared across blocks. Let $\mathbf{x}_t \in \mathbb{R}^C$ be the input token at position t, and let $\mathbf{s}_t \in \mathbb{R}^d$ be the corresponding scale-aware modulation vector from $\mathbf{S}$. The scale-aware state-space update is formulated as

$$\mathbf{h}_t = \mathbf{A}\mathbf{h}_{t-1} + \mathbf{B}\mathbf{x}_t \tag{6}$$

$$\mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{s}_t \odot \mathbf{h}_t + \mathbf{D}\mathbf{x}_t \tag{7}$$

where $\mathbf{h}_t$ and $\mathbf{y}_t$ denote the hidden state and output at position t, respectively, $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are the state-space parameters, and $\odot$ denotes element-wise multiplication. The modulation term $\mathbf{s}_t$ acts as a location-dependent residual adjustment, enabling the model to emphasize scale-sensitive regions, especially complex structures and boundaries.

The modulated sequence embeddings are further refined through state-space propagation to capture long-range dependencies while preserving local scale-sensitive information. They are then projected back to the spatial domain to form the output feature map. In this way, SMEB adapts feature extraction to different target scales and provides robust representations for arbitrary-resolution hyperspectral pan-sharpening.

2) Spatial–Spectral Mamba Interaction Block: To reinforce spatial–spectral interaction during feature extraction, we propose the S^{2}MIB. This module is specifically designed to adaptively integrate high-frequency spatial details from PAN features with the spectral information carried by HSI features.

As illustrated in Fig. 4, the PAN features are first modulated by spectral weights derived from the HSI features to enhance spatial representation in a spectrally guided manner. Meanwhile, a residual connection with the original HSI features is preserved to maintain spectral fidelity. Subsequently, the enhanced HSI features and the depthwise-convolved PAN features are further coupled through the state-space module, enabling more effective spatial–spectral interaction and feature fusion.

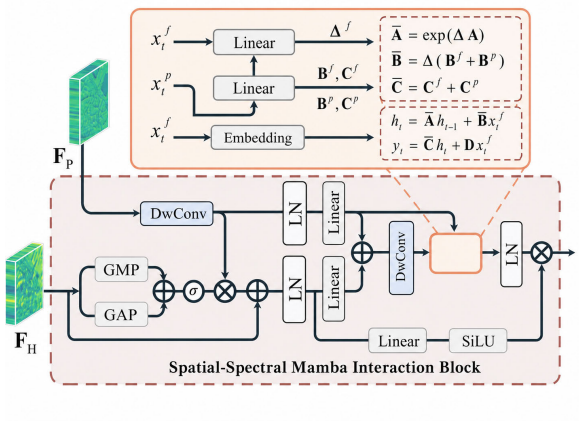


Fig. 4. Implementation details of the S^{2}MIB.

Within the state-space interaction module, the input-to-state and output-related parameters are adaptively generated from two complementary streams: a fused interaction stream and a PAN-guidance stream. Specifically, after projection, the enhanced HSI feature and the PAN feature are split into feature components $\mathbf{x}^h$ and $\mathbf{x}^p$, respectively. The fused interaction stream $\mathbf{x}^f$ is obtained by applying depthwise convolution to the summed HSI and PAN components, while $\mathbf{x}^p$ is retained as the PAN-guidance stream. At each position t, the token $\mathbf{x}_t^f$ is used to generate the fused-stream parameters $\mathbf{\Delta}^f$, $\mathbf{B}^f$, and $\mathbf{C}^f$ through the linear parameter projection $\mathcal{P}(\cdot)$, while $\mathbf{x}_t^p$ is used to generate the PAN-guided parameters $\mathbf{\Delta}^p$, $\mathbf{B}^p$, and $\mathbf{C}^p$ through the same projection. The two-parameter groups are then combined so that the state evolution is jointly conditioned on fused spatial–spectral representations and PAN-guided spatial structures.

The resulting parameters are combined to construct the modulated state-space formulation

$$\bar{\mathbf{A}} = \exp(\mathbf{\Delta}\mathbf{A}) \tag{8}$$

$$\bar{\mathbf{B}} = \mathbf{\Delta}\left(\mathbf{B}^f + \mathbf{B}^p\right) \tag{9}$$

$$\bar{\mathbf{C}} = \mathbf{C}^f + \mathbf{C}^p \tag{10}$$

where $\mathbf{\Delta}$ is obtained by combining the discretization terms from the fused interaction stream and the PAN-guidance stream. Here, $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, and $\bar{\mathbf{C}}$ denote the modulated state transition, input, and output parameters, respectively. The selective scan is performed on the fused interaction stream $\mathbf{x}^f$, while the PAN-guidance stream contributes to spatial-detail-aware parameter modulation. Therefore, the scan output preserves the spectral continuity of the enhanced HSI representation while incorporating PAN-guided spatial details, enabling effective spatial–spectral interaction during feature extraction.

C. Implicit Spatial–Spectral Reconstruction

To enable arbitrary-scale reconstruction of HR-HSIs, we develop the IS^{2}RM, which represents pan-sharpening as a continuous coordinate-based prediction problem. Let

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$\mathcal{H} \in \mathbb{R}^{rh\times rw\times 2}$ denote the target coordinate grid, and let $\mathcal{L} \in \mathbb{R}^{h\times w\times 2}$ denote the low-resolution coordinate grid associated with the interpolated input $\widetilde{\mathbf{Y}}_{\mathrm{H}}$.

Given the extracted HSI and PAN feature maps $\mathbf{F}_{\mathrm{H}}$ and $\mathbf{F}_{\mathrm{P}}$ produced by $\Psi(\widetilde{\mathbf{Y}}_{\mathrm{H}}, \mathbf{Y}_{\mathrm{P}}, r; \theta_E)$, IS^{2}RM predicts the spectral vector at each target query location $\mathbf{c}_q \in \mathcal{H}$ in a query-wise manner. For each $\mathbf{c}_q$, a local supporting window is constructed on $\mathcal{L}$. Let $\mathbf{c}^{(i,j)} \in \mathcal{L}$ denote the supporting coordinate at the (i, j)th position, where $i \in \{1, 2, \ldots, H_w\}$ and $j \in \{1, 2, \ldots, W_w\}$. The relative-coordinate set associated with $\mathbf{c}_q$ is defined as

$$\delta\mathbf{c}_q = \left\{\mathbf{c}_q - \mathbf{c}^{(i,j)}\right\}_{i\in\{1,2,\ldots,H_w\},\, j\in\{1,2,\ldots,W_w\}}. \tag{11}$$

Thus, each query coordinate has $N_w = H_w W_w$ local supporting coordinates, and $\delta\mathbf{c}_q \in \mathbb{R}^{N_w\times 2}$ records their 2-D offsets from the query location.

Based on the extracted features and the query-specific geometric information, the residual spectral vector at $\mathbf{c}_q$ is predicted by

$$\Delta\hat{\mathbf{y}}_q = \Phi\left(\mathbf{F}_{\mathrm{H}}, \mathbf{F}_{\mathrm{P}}, \delta\mathbf{c}_q, \mathbf{s}; \theta_R\right) \tag{12}$$

where $\Delta\hat{\mathbf{y}}_q \in \mathbb{R}^B$ denotes the residual correction to the interpolated LR-HSI value at $\mathbf{c}_q$, and $\mathbf{s} = [s_h, s_w]$ denotes the pixel cell size of the target output grid. Within $\Phi(\cdot)$, $\delta\mathbf{c}_q$ serves two purposes. It provides coordinate-dependent positional bias for local cross-attention and is also used in the local spatial-frequency encoding $\mathcal{E}(\mathbf{F}_{\mathrm{P}}, \delta\mathbf{c}_q)$, enabling PAN-derived high-frequency details to be encoded according to the query-to-support offsets.

To capture both spatial and spectral dependencies, IS^{2}RM adopts two coordinate-conditioned cross-attention branches, namely a spatial branch and a spectral branch. For each branch, the query, key, and value embeddings, denoted by $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, are generated by separate convolutional projections from the extracted feature maps $\mathbf{F}_{\mathrm{H}}$ and $\mathbf{F}_{\mathrm{P}}$.

For each queried coordinate $\mathbf{c}_q$, the query embedding $\mathbf{q}$ is obtained from $\mathbf{Q}$ at $\mathbf{c}_q$, while the key and value embeddings $\mathbf{k}$ and $\mathbf{v}$ are sampled from $\mathbf{K}$ and $\mathbf{V}$ within a local window centered around the nearest low-resolution coordinate. The local sampling set is defined as

$$\mathcal{N}\left(c_q\right) = \left\{c^{(i,j)} \;:\; \left\|c_q - c^{(i,j)}\right\|_\infty \le \rho\right\} \tag{13}$$

where ρ denotes the radius of the local sampling window, and $\mathbf{c}^{(i,j)} \in \mathcal{L}$ are the local supporting coordinates in the low-resolution grid. For a square local window, we have $H_w = W_w = 2\rho + 1$ and $N_w = H_w W_w = (2\rho + 1)^2$.

In the spatial branch, the branch-specific query embedding is denoted as $\mathbf{q}_{\mathrm{spa}} \in \mathbb{R}^{1\times C}$, and the local key and value embeddings sampled from $\mathcal{N}(c_q)$ are denoted as $\mathbf{k}_{\mathrm{spa}} \in \mathbb{R}^{N_w\times C}$ and $\mathbf{v}_{\mathrm{spa}} \in \mathbb{R}^{N_w\times C}$, respectively, where C is the channel dimension. To distinguish different local supporting positions, a coordinate-dependent positional bias is generated from $\delta\mathbf{c}_q$

$$\mathbf{B}_{\mathrm{pos}} = \mathrm{FC}\left(\varphi\left(\delta\mathbf{c}_q\right)\right) \tag{14}$$

where $\varphi(\cdot)$ is the periodic encoding function and $\mathbf{B}_{\mathrm{pos}} \in \mathbb{R}^{1\times N_w}$. The spatial attention output is then computed as

$$\mathbf{z}_{\mathrm{spa}} = \mathrm{Softmax}\left(\frac{\mathbf{q}_{\mathrm{spa}}\mathbf{k}_{\mathrm{spa}}^{\top}}{\sqrt{C}} + \mathbf{B}_{\mathrm{pos}}\right)\mathbf{v}_{\mathrm{spa}}. \tag{15}$$

Since $\mathbf{q}_{\mathrm{spa}}\mathbf{k}_{\mathrm{spa}}^{\top} \in \mathbb{R}^{1\times N_w}$, the Softmax operation is applied over the N_w local supporting positions, yielding $\mathbf{z}_{\mathrm{spa}} \in \mathbb{R}^{1\times C}$.

The spectral branch follows the same procedure with branch-specific query, key, and value embeddings, but exchanges the roles of the two modalities to obtain the latent spectral embedding $\mathbf{z}_{\mathrm{spe}}$. The two branch outputs are then adaptively aggregated as

$$\mathbf{z}'_{\mathrm{spa}} = \mathbf{z}_{\mathrm{spa}} \odot \mathbf{w}_{\mathrm{spe}} + \mathbf{z}_{\mathrm{spa}} \tag{16}$$

$$\mathbf{z}'_{\mathrm{spe}} = \mathbf{z}_{\mathrm{spe}} \odot \mathbf{w}_{\mathrm{spa}} + \mathbf{z}_{\mathrm{spe}} \tag{17}$$

$$\tilde{\mathbf{z}} = \mathbf{z}'_{\mathrm{spa}} + \mathbf{z}'_{\mathrm{spe}} \tag{18}$$

where $\mathbf{w}_{\mathrm{spe}}$ and $\mathbf{w}_{\mathrm{spa}}$ denote branch-wise attention weights generated from global pooling and convolutional operations, and $\odot$ denotes element-wise multiplication. Specifically, $\mathbf{w}_{\mathrm{spe}}$ is generated from the spectral branch to recalibrate $\mathbf{z}_{\mathrm{spa}}$, while $\mathbf{w}_{\mathrm{spa}}$ is generated from the spatial branch to recalibrate $\mathbf{z}_{\mathrm{spe}}$. Both weights have the same channel dimension as the corresponding latent embeddings and are broadcast during element-wise multiplication.

To mitigate the frequency bias of INRs, a local spatial-frequency encoding is introduced by modulating PAN-derived local frequency features with periodic relative-coordinate encoding

$$\mathcal{E}\left(\mathbf{F}_{\mathrm{P}}, \delta\mathbf{c}_q\right) = \mathcal{F}\left(\mathcal{N}_{c_q}\left(\mathrm{Conv}\left(\mathbf{F}_{\mathrm{P}}\right)\right)\right) \odot \varphi\left(\delta\mathbf{c}_q\right) \tag{19}$$

where $\mathcal{N}_{c_q}(\cdot)$ denotes the local sampling operation centered at the queried coordinate c_q, $\mathcal{F}(\cdot)$ denotes the fast Fourier transform, and $\varphi(\cdot)$ is the periodic positional encoding function.

Finally, the refined latent embedding $\tilde{\mathbf{z}}$, the local spatial-frequency encoding $\mathcal{E}(\mathbf{F}_{\mathrm{P}}, \delta\mathbf{c}_q)$, and the target pixel size $\mathbf{s}$ are concatenated and fed into a five-layer MLP decoder D_ϕ with GeLU activations to predict the residual spectral vector

$$\Delta\hat{\mathbf{y}}_q = D_\phi\left(\mathrm{Concat}\left(\tilde{\mathbf{z}}, \mathcal{E}\left(\mathbf{F}_{\mathrm{P}}, \delta\mathbf{c}_q\right), \mathbf{s}\right)\right). \tag{20}$$

This equation gives the residual prediction step inside $\Phi(\cdot)$, while the complete reconstruction module also includes coordinate-dependent positional modeling and dual-branch latent feature aggregation. Following the common residual prediction strategy in implicit reconstruction, the final spectral prediction at coordinate c_q is obtained by adding the residual decoder output to the corresponding pixel value sampled from the interpolated LR-HIS

$$\hat{\mathbf{y}}_q^{\mathrm{final}} = \widetilde{\mathbf{Y}}_{\mathrm{H}}\left(c_q\right) + \Delta\hat{\mathbf{y}}_q. \tag{21}$$

IV. Experiment

A. Datasets and Benchmarks

To evaluate the effectiveness of the proposed approach, we conduct comprehensive experiments on three widely used benchmark datasets.

1) Pavia Center: Pavia Center is a representative urban hyperspectral benchmark acquired over Pavia, Italy. The effective scene used in this study contains 1096×715 pixels with 102 spectral bands after preprocessing. Its clear boundaries and rich structural details make it suitable for evaluating spatial sharpness and spectral fidelity.

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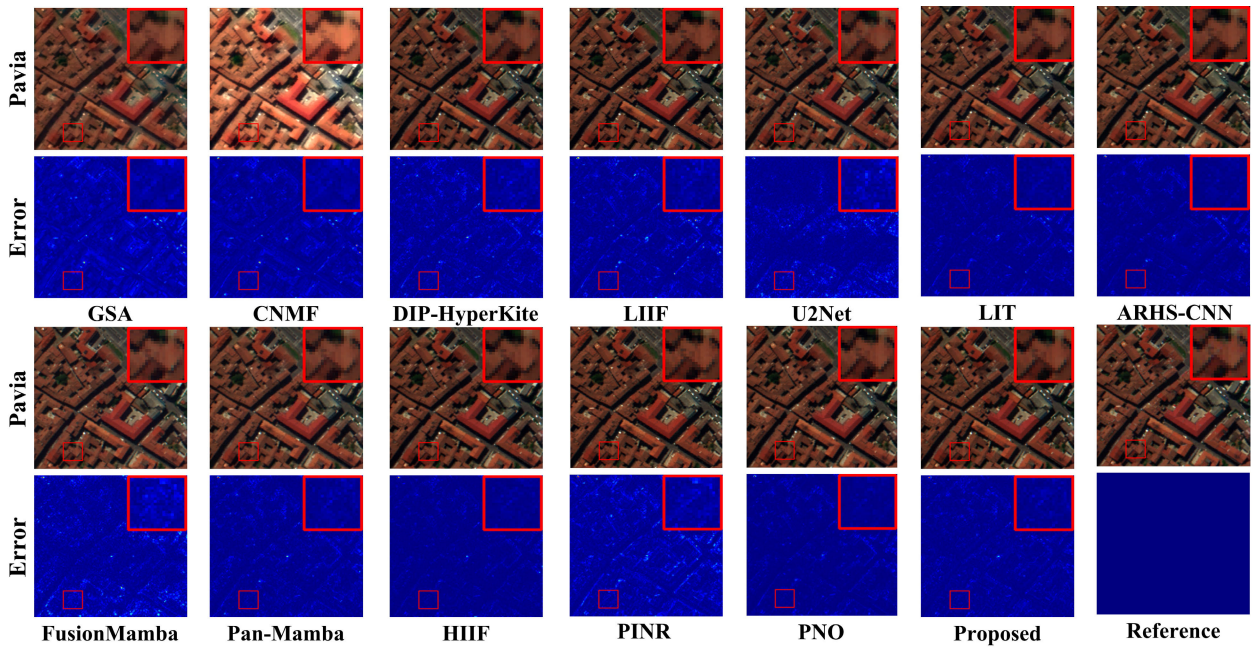


Fig. 5. Qualitative results of standard pan-sharpening on Pavia Center dataset.

2) Chikusei: Chikusei is a large-scale airborne hyperspectral dataset collected over Chikusei, Japan. The original scene consists of 2517 × 2335 pixels with 128 spectral bands, covering agricultural and suburban areas. Its homogeneous fields, road networks, and scattered structures provide multiscale textures for evaluating scale adaptation and generalization.

3) Houston: Houston is a challenging urban hyperspectral benchmark with complex man-made structures and heterogeneous land-cover patterns. The original scene spans 349×1905 pixels with 144 spectral bands. Its dense details and diverse regions make it suitable for evaluating fine-structure recovery and spectral distortion control.

Since real HR-HSI references are generally unavailable in practical acquisitions, all experiments are conducted on simulated data following Wald's protocol [64], [65], [66]. The original HR-HSI is used as the ground truth, while the corresponding PAN image and LR-HSI are generated in a controlled manner, ensuring that all competing methods are evaluated with consistent inputs and the same reference. Specifically, the PAN image is synthesized by averaging the HSI bands within the spectral coverage of the PAN sensor to preserve high-resolution spatial structures. The LR-HSI is generated band by band from the reference HR-HSI. For each spectral band, a Gaussian low-pass filter with a kernel size of 8 × 8 and circular boundary padding is first applied, and the blurred result is then downsampled by bicubic interpolation with the fixed spatial ratio $r_0 = 4$. The same degradation kernel and downsampling process are applied to all spectral bands to ensure spectral consistency and spatial registration among the simulated PAN image, LR-HSI, and reference HR-HSI.

For comprehensive evaluation, we compare our method with representative traditional and deep learning-based pan-sharpening methods, including GSA [27], CNMF [31], DIP-HyperKite [34], LIIF [49], LIT [52], HIIF [53], U2Net [58], FusionMamba [42], Pan-Mamba [43], ARHS-CNN [45], PINR [54], and PNO [62]. Baselines with official implementations are retrained on the same data, while ARHS-CNN, PINR, and PNO are reproduced following their original papers. All methods follow the same data setting and preprocessing pipeline for fair comparison. For LIIF, HIIF, and LIT, we concatenate LR-HSI and PAN inputs and use the same scale-factor setting for arbitrary-resolution pan-sharpening. We adopt ERGAS [67], SAM [68], Q2n [69], and PSNR [69] for evaluation. Lower ERGAS/SAM and higher Q2n/PSNR indicate better spatial–spectral reconstruction under the simulated protocol.

B. Implementation Details

All experiments are conducted using PyTorch on Ubuntu 18.04 with a single NVIDIA GeForce RTX 3090 GPU with 24 GB VRAM. The network is optimized by Adam with the L1 loss. The initial learning rate is set to 2×10^{-4} for Pavia Center and Chikusei, and 1×10^{-4} for Houston. The learning rate is decayed by a factor of 0.5 every 200 epochs, and the model is trained for 1000 epochs to ensure convergence.

During training, PAN, LR-HSI, and HR-HSI patches are cropped in a scale-consistent manner. The LR-HSI input size is set to 40 × 40, and the PAN input is resized to the same spatial size before being fed into the network. For each mini-batch, the PAN-referenced scale factor r' is randomly sampled from the predefined training scale range. The target HR-HSI and PAN patches are cropped with size $40r' \times 40r'$ at the same top-left coordinate, while the corresponding LR-HSI crop is extracted from the low-resolution grid according to the fixed ratio r_0. The cropped PAN and LR-HSI patches are resized to 40 × 40 as network inputs, and the HR-HSI crop of size $40r' \times 40r'$ is used as supervision.

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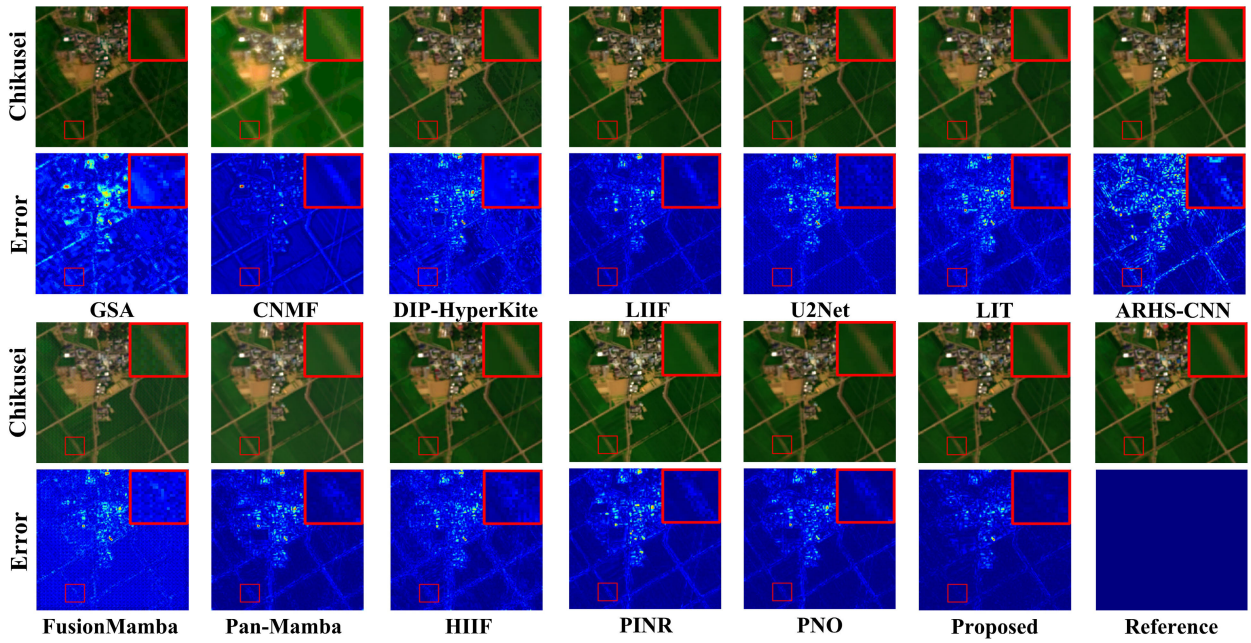


Fig. 6. Qualitative results of standard pan-sharpening on Chikusei dataset.

A continuous coordinate grid is then constructed at the target resolution for scale-conditioned pixel-wise prediction. To reduce the training cost of coordinate-based implicit reconstruction, 1600 query coordinates are randomly sampled from the target coordinate grid for each training sample. The target cell size is computed according to the target output resolution and used as an additional input to the implicit decoder.

During inference, the output resolution is determined by the desired scale factor. Given the target height H_{out} and width W_{out}, we generate the full coordinate grid and compute the cell size as $\mathbf{s} = [2/H_{\text{out}}, 2/W_{\text{out}}]$. The model predicts the hyperspectral value at each queried coordinate. To reduce memory consumption, the full set of query coordinates is split into smaller chunks and processed by chunked coordinate querying. The default inference query batch size is set to 10 000 in the reconstruction module. The predicted values are then reshaped to the target HR-HSI size and clipped to the range $[0, 1]$.

During testing, we vary the PAN-referenced scale factor r' to generate pan-sharpened outputs at different spatial sizes. Since $r = r_0 r'$ and $r_0 = 4$ in our experiments, this is equivalent to varying the overall LR-HSI-to-output upsampling factor r from 4 to 16. This process only changes the query coordinate grid and the associated cell size, enabling continuous-scale prediction beyond the training scales without changing the learned parameters.

C. Comparison on Standard Pan-Sharpening

To rigorously evaluate the proposed method, we evaluate the proposed method against representative baselines under a unified standard pan-sharpening setting.

For arbitrary-resolution methods, the output scale factor is fixed at $r' = 1$ during inference so that all methods are evaluated under the same standard pan-sharpening setting and target resolution. Tables I–III report ERGAS, SAM, Q2n, and PSNR on Pavia Center, Chikusei, and Houston, enabling a consistent comparison across datasets with different scene characteristics. Overall, the proposed method achieves the lowest ERGAS and SAM and the highest PSNR on all three datasets, while remaining competitive in terms of Q2n.

Figs. 5–7 show visual comparisons and residual maps. Traditional methods tend to suffer from spectral distortion or over-smoothing, while some deep baselines still produce artifacts and incomplete fine details. In contrast, the proposed method preserves sharper edges and clearer textures, with darker residual maps indicating smaller reconstruction errors.

TABLE I

QUALITY METRICS OF STANDARD PAN-SHARPENING ON THE PAVIA CENTER. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED

Methods	ERGAS	SAM	Q2n	PSNR
GSA	5.1301	7.6309	0.9595	32.0495
CNMF	4.7830	7.0503	0.9652	32.6615
DIP-HyperKite	2.7291	4.9062	0.9921	38.6274
LIIF	3.1095	4.9596	0.9889	37.0277
U2Net	<u>2.4491</u>	<u>4.3039</u>	0.9935	39.7436
LIT	3.0220	5.0048	0.9896	37.4348
FusionMamba	2.5323	4.4582	0.9930	39.3328
Pan-Mamba	2.4947	4.6268	<u>0.9940</u>	<u>39.7969</u>
HIIF	2.4981	4.4176	0.9934	39.4149
ARHS-CNN	2.9902	5.0759	0.9898	37.5690
PINR	3.3279	5.2652	0.9867	36.2272
PNO	3.0505	4.5184	0.9901	39.7604
SAMIR	**2.3852**	**4.2863**	**0.9942**	**40.0849**
Ideal value	0	0	1	$+\infty$

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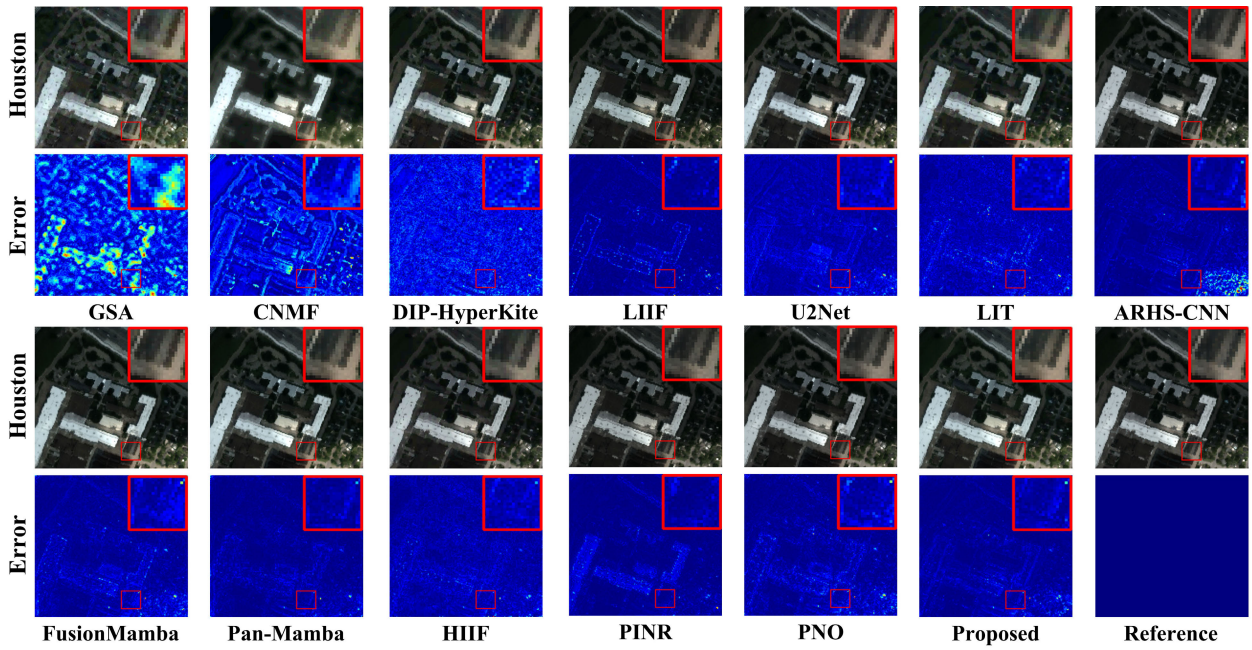


Fig. 7. Qualitative results of standard pan-sharpening on Houston dataset.

TABLE II

QUALITY METRICS OF STANDARD PAN-SHARPENING ON THE CHIKUSEI DATASET. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED

Methods	ERGAS	SAM	Q2n	PSNR
GSA	5.4036	3.7294	0.9685	40.1740
CNMF	5.9580	3.7981	0.9698	39.5301
DIP-HyperKite	3.9097	2.6411	0.9831	43.3859
LIIF	3.4761	2.5404	0.9836	43.4251
U2Net	3.0126	2.2825	0.9887	45.0628
LIT	3.1958	2.4440	0.9860	44.0163
FusionMamba	4.0739	2.9591	0.9865	42.4387
Pan-Mamba	2.7963	2.1350	0.9896	45.8122
HIIF	2.7977	2.1468	<u>0.9904</u>	45.5719
ARHS-CNN	3.1879	2.3534	0.9877	44.9643
PINR	3.2496	2.4083	0.9844	43.8393
PNO	<u>2.7540</u>	<u>1.9786</u>	0.9868	<u>47.2436</u>
SAMIR	**2.3832**	**1.8044**	**0.9921**	**47.6310**
Ideal value	0	0	1	$+\infty$

TABLE III

QUALITY METRICS OF STANDARD PAN-SHARPENING ON THE HOUSTON DATASET. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED

Methods	ERGAS	SAM	Q2n	PSNR
GSA	4.7877	5.6633	0.9661	38.8844
CNMF	4.8306	5.8586	0.9656	38.8744
DIP-HyperKite	3.2162	3.8812	0.9844	43.7594
LIIF	2.9821	3.4879	0.9827	41.9687
U2Net	<u>2.1284</u>	2.7803	**0.9917**	<u>45.9697</u>
LIT	2.7736	3.5280	0.9852	43.4175
FusionMamba	2.2628	<u>2.7601</u>	<u>0.9914</u>	45.5174
Pan-Mamba	2.2661	2.9508	0.9896	45.5100
HIIF	2.4162	3.1333	0.9890	44.4969
ARHS-CNN	3.0107	3.8596	0.9833	42.9998
PINR	3.0727	3.6717	0.9818	42.1729
PNO	2.7747	3.0737	0.9752	43.1624
SAMIR	**2.1218**	**2.6468**	0.9909	**46.2421**
Ideal value	0	0	1	$+\infty$

Overall, the quantitative and qualitative results demonstrate the effectiveness of the proposed method under the standard pan-sharpening setting. It achieves lower spectral distortion, better spatial detail preservation, and competitive spatial–spectral consistency across different datasets.

D. Comparison on Arbitrary-Scale Pan-Sharpening

To evaluate scale-adaptive robustness, we conduct arbitrary-resolution experiments on three benchmark datasets by sweeping the PAN-referenced output scale ratio r' from 1.0 to 4.0 with a step size of 0.05. Since $r = r_0 r'$ and $r_0 = 4$, this corresponds to varying the overall LR-HSI-to-output upsampling factor r from 4 to 16, ranging from standard pan-sharpening to 16× reconstruction. The training range is $r' \in [1, 2]$, corresponding to $r \in [4, 8]$. Therefore, the gray region in Fig. 8 denotes the interpolation range, while the white region denotes the extrapolation range, i.e., $r' \in (2, 4]$ or equivalently $r \in (8, 16]$. For $r' > 1$, the output exceeds the PAN sampling resolution and is interpreted as PAN-guided hyperspectral spatial super-resolution, where spectral content is constrained by LR-HSI and sub-PAN spatial details are inferred from PAN structures and learned spatial–spectral priors.

To evaluate scale-adaptive robustness, we sweep the PAN-referenced scale ratio r' from 1.0 to 4.0 with a step size of 0.05 on three benchmark datasets. Since $r = r_0 r'$ and $r_0 = 4$, the overall LR-HSI-to-output scale r ranges from 4 to 16. The training range is $r' \in [1, 2]$ ($r \in [4, 8]$), marked as the gray interpolation region in Fig. 8, while $r' \in (2, 4]$ ($r \in (8, 16]$)

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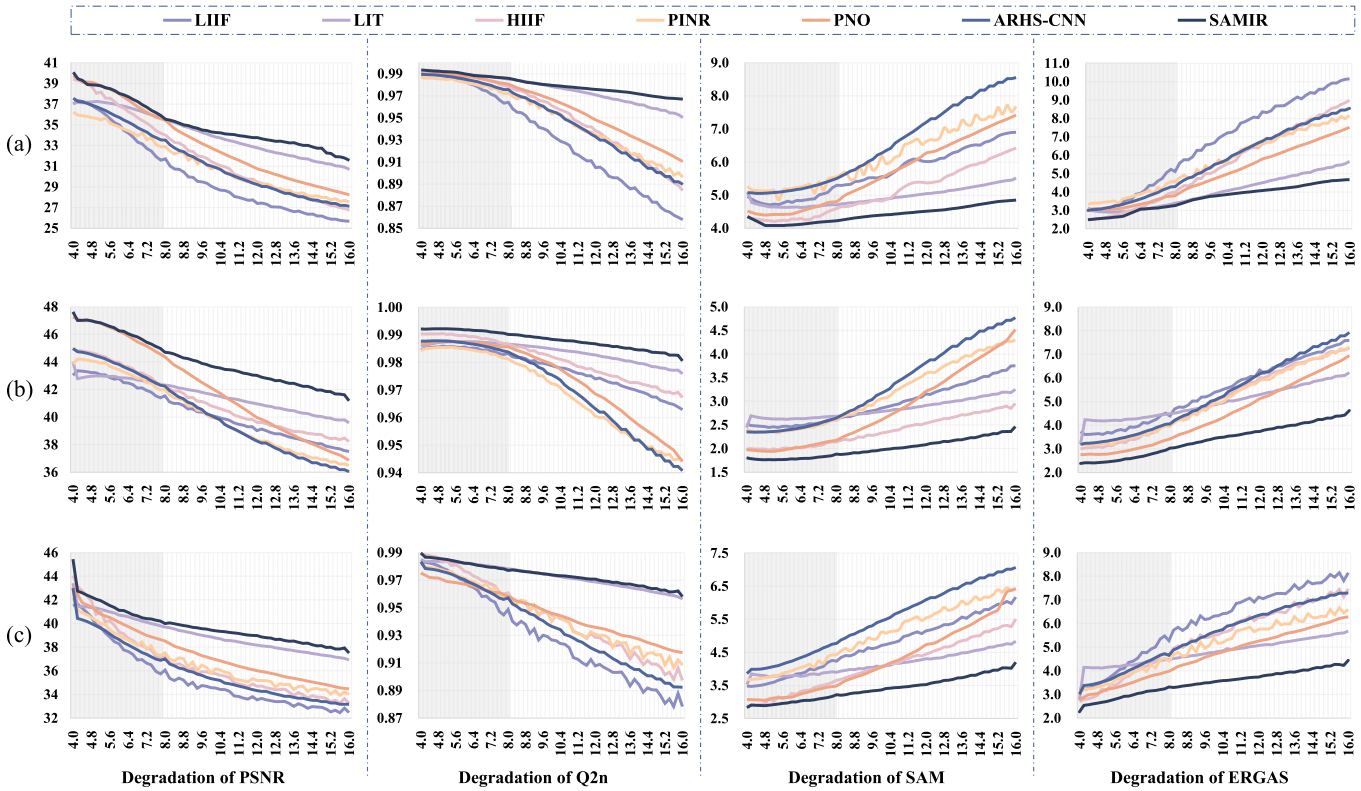


Fig. 8. Degradation trends of PSNR, Q2n, SAM, and ERGAS with increasing overall upsampling factor r (from 4 to 16) on (a) Pavia Center, (b) Chikusei, and (c) Houston, comparing LIIF, LIT, HIIF, ARHS-CNN, PINR, PNO, and SAMIR-Net. Here, r denotes the overall LR-HSI-to-output scale and satisfies $r = r_0 r'$; under our experimental setting $r_0 = 4$, the plotted range $r \in [4, 16]$ corresponds to $r' \in [1.0, 4.0]$.

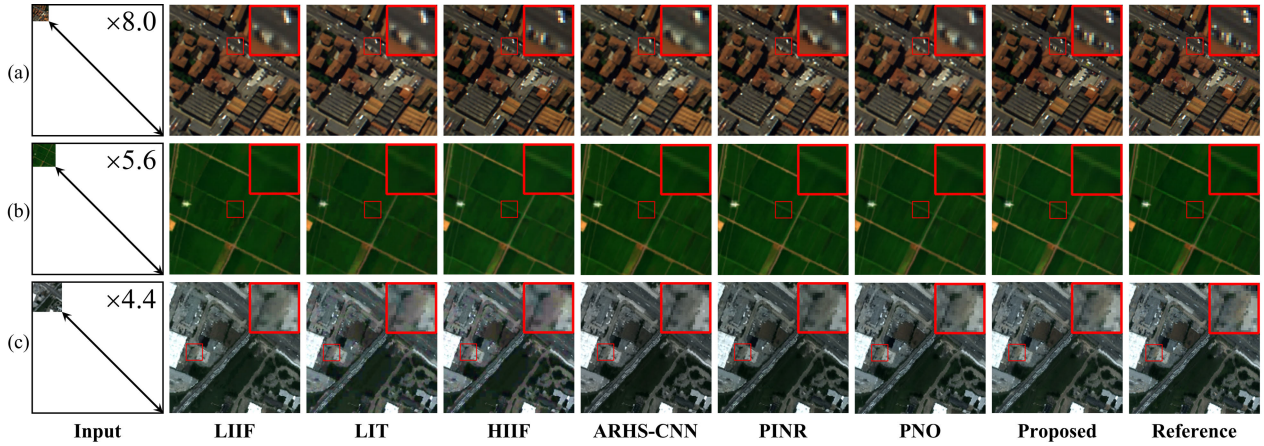


Fig. 9. Visual comparison of arbitrary-scale pan-sharpening results on Pavia, Chikusei, and Houston (top to bottom) at overall scales $r = 8.0$, 5.6, and 4.4 (corresponding to $r' = 2.0$, 1.4, and 1.1 under $r_0 = 4$).

is the white extrapolation region. For $r' > 1$, the output exceeds the PAN resolution and is regarded as PAN-guided hyperspectral spatial super-resolution, where sub-PAN details are inferred from PAN structures and learned spatial–spectral priors.

As shown in Fig. 8, all methods degrade as the target scale increases, with decreasing PSNR/Q2n and increasing SAM/ERGAS. Table IV further reports the average results in the interpolation and extrapolation ranges. SAMIR-Net achieves the best average performance on all three datasets in both ranges. In the extrapolation range, it reaches PSNR values of 33.6296, 42.9706, and 38.8779 dB on Pavia Center, Chikusei, and Houston, outperforming the strongest competitors by 0.79, 2.03, and 0.68 dB, respectively. It also obtains the best ERGAS, SAM, and Q2n, demonstrating stronger robustness to unseen larger scales.

The visual comparisons in Fig. 9 further support the quantitative results. Representative examples at $r = 8.0$, 5.6, and 4.4 are shown for Pavia Center, Chikusei, and Houston, respectively. Existing methods tend to produce softened boundaries or local structural artifacts in the highlighted regions, whereas SAMIR-Net recovers clearer edges and more faithful fine structures. These visual results confirm that the proposed method preserves local structures

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TABLE IV

AVERAGE QUALITY METRICS IN THE INTERPOLATION AND EXTRAPOLATION RANGES. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED

Dataset	Method	In-distribution				Out-of-distribution			
		ERGAS	SAM	Q2n	PSNR	ERGAS	SAM	Q2n	PSNR
Pavia	LIIF	3.8509	4.8872	0.9803	34.7449	8.1340	6.0751	0.9068	27.7738
	LIT	3.0616	4.6921	0.9887	36.6782	4.5270	5.0736	0.9697	32.8386
	HIIF	3.0929	4.3193	0.9880	36.8354	6.6222	5.4392	0.9353	29.6926
	ARHS-CNN	3.5479	5.1964	0.9841	35.5984	6.7030	7.1684	0.9319	29.4971
	PINR	3.8102	5.2257	0.9813	34.7027	6.5722	6.6422	0.9340	29.6007
	PNO	3.3157	4.5387	0.9869	37.7130	5.7688	6.2115	0.9471	31.0506
	SAMIR	**2.8667**	**4.1588**	**0.9899**	**37.8861**	**4.0955**	**4.5427**	**0.9756**	**33.6296**
Chikusei	LIIF	3.9195	2.5054	0.9849	42.5608	6.1786	3.1583	0.9738	39.1934
	LIT	4.2356	2.6389	0.9871	42.8221	5.3283	2.9314	0.9821	40.9393
	HIIF	3.3926	2.0304	0.9893	43.8005	5.8865	2.5444	0.9769	39.8470
	ARHS-CNN	3.5519	2.4396	0.9866	43.6963	6.1247	3.7845	0.9627	38.5223
	PINR	3.4618	2.4262	0.9843	43.2931	5.8022	3.5345	0.9615	38.6269
	PNO	2.9738	2.0263	0.9869	46.0502	5.1468	3.2478	0.9674	40.1684
	SAMIR	**2.6077**	**1.7950**	**0.9917**	**46.2205**	**3.7789**	**2.1096**	**0.9864**	**42.9706**
Houston	LIIF	4.2046	3.8027	0.9675	38.4732	6.9993	5.1926	0.9092	33.7544
	LIT	4.1903	3.8077	0.9820	40.8072	5.0780	4.3243	0.9682	38.2021
	HIIF	3.6813	3.2273	0.9761	39.7906	6.2342	4.5218	0.9278	34.8063
	ARHS-CNN	3.9493	4.2857	0.9692	38.7951	6.2689	6.0532	0.9200	34.4978
	PINR	3.7634	3.9732	0.9709	39.2335	5.7183	5.5604	0.9306	35.2941
	PNO	3.4095	3.2218	0.9666	40.2570	5.2419	4.7840	0.9361	36.1152
	SAMIR	**2.8821**	**3.0018**	**0.9827**	**41.4454**	**3.7925**	**3.6002**	**0.9696**	**38.8779**

TABLE V

QUANTITATIVE ABLATION RESULTS ON THE PAVIA CENTER DATASET. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED

Methods	ERGAS	SAM	Q2n	PSNR
w/o Scale-aware	2.5832	4.4739	0.9926	38.9872
w/o Interaction	2.5970	**4.1535**	0.9921	38.4252
w/o dual-branch	2.5556	4.4565	0.9930	39.1656
SAMIR	**2.3852**	4.2863	**0.9942**	**40.0849**
Ideal value	0	0	1	$+\infty$

and visual fidelity more effectively under different target scales.

Overall, both quantitative and qualitative results show that SAMIR-Net achieves slower performance degradation, better spatial detail recovery, and stronger spectral preservation across interpolation and extrapolation ranges, confirming its effectiveness for arbitrary-resolution hyperspectral pan-sharpening. Nevertheless, these arbitrary-scale results are based on the reduced-resolution Wald protocol. They demonstrate scale generalization in a controlled reference-based setting, but cannot replace a real full-resolution evaluation. Further validation with real multisensor data, no-reference metrics, and downstream tasks is needed for practical assessment.

E. Ablation Study

1) Effect of Core Components: To assess the contribution of each core module, we conduct ablation experiments by individually removing the scale-aware term in SASSM, the S^{2}MIB, and the dual-branch architecture in IS^{2}RM. This design allows us to isolate the effect of each component and analyze how it contributes to the overall reconstruction quality from both spatial and spectral perspectives.

We evaluate the contribution of three core designs by removing the scale-aware term in SASSM, the S^{2}MIB, and the dual-branch architecture in IS^{2}RM, respectively. As shown in Table V, removing any component degrades the overall performance, confirming that scale-aware modeling, cross-modal interaction, and dual-branch reconstruction are all beneficial to the proposed framework.

TABLE VI

MASK-BASED EVALUATION ON SCALE-SENSITIVE REGIONS. FOR PSNR, A SMALLER RELATIVE DEGRADATION INDICATES BETTER ROBUSTNESS; FOR SAM, A SMALLER OR NEGATIVE CHANGE INDICATES LESS SPECTRAL DEGRADATION

Method	Unmasked		Masked		Relative degradation (%)	
	PSNR↑	SAM↓	PSNR↑	SAM↓	PSNR↓	SAM↓
SAMIR	**39.24**	**2.96**	**37.89**	**2.85**	**3.44**	**3.72**
LIIF	36.82	3.71	35.03	3.91	4.86	5.39
HIIF	38.86	3.06	37.44	3.23	3.65	5.56
LIT	36.53	3.91	34.90	4.20	4.46	7.42
ARHS-CNN	34.75	5.09	31.76	5.34	8.34	4.59
PINR	34.37	4.17	31.75	4.60	7.65	10.38
PNO	36.70	3.33	34.55	3.60	5.78	7.80

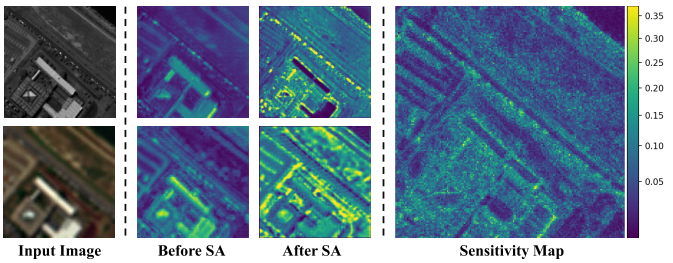


Fig. 10. Visualization of the feature maps before and after scale-aware conv layer and the corresponding scale sensitivity map.

Removing the scale-aware term causes a clear performance drop, indicating the importance of explicit scale modeling for arbitrary-resolution pan-sharpening. As shown in Fig. 10, the scale-aware operation enhances responses in fine structures and scale-sensitive regions, thereby improving spatial detail preservation under varying target scales. To further evaluate robustness in scale-sensitive regions, we conduct a mask-based analysis using the pixel-wise AUC map as Fig. 3. The unmasked results reflect overall reconstruction quality, while the masked results focus on highly scale-sensitive pixels. As shown in Table VI, competing methods suffer larger degradation in the masked regions, whereas SAMIR-Net shows the smallest PSNR decrease and no SAM increase under mask weighting. These results confirm that the proposed scale-aware design improves robustness in regions vulnerable to scale variation.

Eliminating the S^{2}MIB also degrades reconstruction quality, especially in spatial detail recovery. This indicates that effective interaction between PAN-derived spatial cues and HSI spectral features is essential for cross-modal fusion.

Excluding the dual-branch design in IS^{2}RM further reduces performance, confirming the need to separately model spatial and spectral dependencies before final aggregation. Overall, these results show that scale-aware modeling, cross-modal interaction, and dual-branch reconstruction jointly contribute to the best balance between spatial enhancement and spectral consistency.

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TABLE VII

PERFORMANCE IMPACT OF VARYING THE NUMBER OF LAYERS L IN THE SCALE-AWARE MODALITY INTERACTIVE EXTRACTION ON THE PAVIA CENTER DATASET. BEST RESULTS ARE IN BOLD, AND SECOND-BEST ARE UNDERLINED

L	ERGAS	SAM	Q2n	PSNR
2	2.6083	4.5089	0.9924	38.8802
3	2.4999	4.3456	0.9933	39.3918
4	**2.3852**	4.2863	**0.9942**	**40.0849**
5	2.3944	**4.2197**	0.9941	39.9485
6	2.3910	4.2301	0.9941	39.9522
Ideal value	0	0	1	$+\infty$

TABLE VIII

OBSERVATION-CONSISTENCY VALIDATION FOR BEYOND-PAN RECONSTRUCTION. RESULTS ARE AVERAGED OVER $r' \in \{1.5, 2.0, 3.0, 4.0\}$. LIT AND PNO ARE REPORTED AS THE BEST AND SECOND-BEST REPRESENTATIVE ARBITRARY-RESOLUTION BASELINES, RESPECTIVELY

Dataset	Metric	LIT	PNO	SAMIR
Pavia	$SAM_{\mathrm{LR}} \downarrow$	3.9021	4.0101	**3.4099**
	$CC_{\mathrm{PAN}} \uparrow$	0.99737	0.98835	**0.99945**
Chikusei	$SAM_{\mathrm{LR}} \downarrow$	2.1298	2.4157	**1.8765**
	$CC_{\mathrm{PAN}} \uparrow$	0.98730	0.98022	**0.98951**
Houston	$SAM_{\mathrm{LR}} \downarrow$	4.5440	4.7130	**3.5433**
	$CC_{\mathrm{PAN}} \uparrow$	0.97366	0.95973	**0.99743**

2) Influence of Layer Number: We further study the influence of the layer number L in the scale-aware modality interactive extraction module. As reported in Table VII, performance improves as L increases to 4, indicating that moderate depth helps capture more effective cross-modal representations. When L is further increased, the gains become marginal and a slight degradation appears on some metrics, suggesting possible redundancy and increased optimization difficulty. Therefore, we set $L = 4$ in the final model to balance reconstruction accuracy and computational efficiency.

3) Validation of Beyond-Pan Reconstruction: To further validate the beyond-PAN setting, we conduct an observation-consistency evaluation for the outputs with $r' > 1$. The reconstructed HR-HSI is degraded back to the original LR-HSI resolution using the same degradation process as adopted in the simulated data generation, and then compared with the input LR-HSI using SAM_{LR}. Meanwhile, a PAN image is synthesized from the reconstructed HSI using the same PAN synthesis strategy as in the dataset construction, resized to the observed PAN resolution, and compared with the input PAN using CC_{PAN}. As shown in Table VIII, we report LIT and PNO as the best and second-best representative arbitrary-resolution baselines, respectively. SAMIR-Net achieves the lowest SAM_{LR} and the highest CC_{PAN} on all three datasets, indicating that its beyond-PAN outputs remain more consistent with the observable LR-HSI and PAN constraints. This suggests that the inferred details are guided by available observations and learned spatial–spectral priors rather than unconstrained hallucinations.

4) Accuracy and Computational Cost Assessment: Fig. 11 compares the accuracy–efficiency tradeoff of arbitrary-resolution pan-sharpening methods in terms of parameters, reconstruction accuracy, and FLOPs. SAMIR-Net achieves the highest PSNR and lowest SAM among all methods. Compared with LIIF, LIT, and HIIF, it obtains better reconstruction quality with fewer parameters, showing the effectiveness of the proposed scale-aware spatial–spectral interaction design. Although PINR is more lightweight and PNO has lower FLOPs, both show inferior reconstruction accuracy. These results indicate that SAMIR-Net provides a favorable balance between accuracy and model complexity.

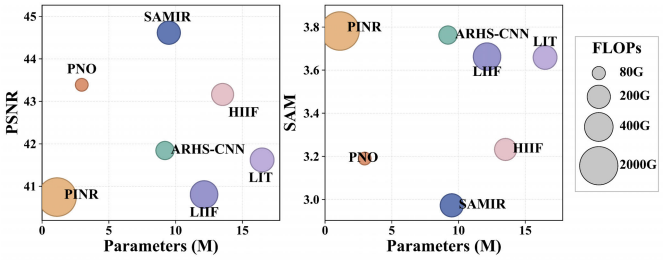


Fig. 11. Comparison of our method with arbitrary-resolution baselines in terms of computational complexity.

TABLE IX

PEAK MEMORY USAGE AND INFERENCE TIME OF ARBITRARY-RESOLUTION METHODS UNDER DIFFERENT QUERY CHUNK SIZES. MEM. AND TIME DENOTE MEMORY USAGE (MB) AND INFERENCE TIME (MS), RESPECTIVELY

Method	$N_q = 5{,}000$		$N_q = 10{,}000$		$N_q = 20{,}000$	
	Mem.	Time	Mem.	Time	Mem.	Time
LIIF	876	117.93	954	112.27	1014	112.15
LIT	2468	257.23	4088	205.63	7346	221.80
HIIF	858	126.41	900	122.66	1100	124.84
PINR	432	98.55	452	88.44	492	82.46
SAMIR	2932	116.80	4226	106.09	7990	107.85

Table IX reports peak memory and inference time under different query chunk sizes (N_q = 5000, 10 000, and 20 000). For INR-based methods, N_q affects memory usage due to chunked coordinate querying. As shown in Table IX, larger N_q generally increases peak memory, while runtime varies with query-splitting overhead, memory access, and parallel efficiency. SAMIR-Net achieves competitive inference speed, consistently outperforming LIT and remaining comparable to LIIF and HIIF. Although PINR is faster, its accuracy is noticeably lower, as shown in Fig. 11.

Overall, these results reveal different tradeoffs among accuracy, model size, computational cost, memory usage, and inference time. SAMIR-Net achieves the best reconstruction accuracy with a moderate parameter scale and competitive speed, offering a favorable accuracy–efficiency balance for arbitrary-resolution hyperspectral pan-sharpening. Its relatively high peak memory usage also suggests that reducing the memory footprint of query-based reconstruction remains an important direction for future optimization.

V. CONCLUSION

In this article, we propose a scale-aware Mamba-driven framework for arbitrary-resolution hyperspectral pan-sharpening. By incorporating scale priors into feature extrac-

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tion and reconstruction, the proposed method enhances PAN-guided spatial detail injection while preserving HSI spectral information across different target resolutions. Experiments on three benchmark datasets show that the proposed method achieves competitive performance under the standard pan-sharpening setting and maintains relatively stable reconstruction quality under arbitrary-resolution settings. These results indicate the potential of SAMIR-Net for flexible single-model hyperspectral pan-sharpening. In future work, we will improve the framework from three aspects: first, reducing the memory cost of query-wise reconstruction. Second, developing registration-aware designs for misaligned LR-HSI/PAN inputs, and exploring more reliable evaluation protocols for practical and beyond-PAN settings. Third, considering that evaluation remains a key challenge in hyperspectral pan-sharpening, we will explore no-reference and task-driven evaluation for standard full-resolution settings, and further investigate observation-consistency and cross-scale consistency validation for beyond-PAN reconstruction, where conventional no-reference metrics are not directly applicable.

REFERENCES

[1] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, 2022, Art. no. 152102.

[2] S. Mei et al., "Lightweight multiresolution feature fusion network for spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501414.

[3] C. Sui, Z. Xu, Y. Meng, H. Wang, B. Zhang, and G. Vivone, "CycleMamba: Cycle-consistent learning for aerial visible-to-infrared image translation," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5606818.

[4] C. Sui et al., "IG-GAN: Interactive guided generative adversarial networks for multimodal image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5634719.

[5] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[6] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[7] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[8] W. Liu, H. Bi, X. Zhu, C. Xu, and B. Pan, "Unsupervised and unpaired fusion-based hyperspectral image super-resolution based on reflectance migration," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5516615.

[9] P. Wang et al., "CIRSM-Net: A cyclic registration network for SAR and optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610619.

[10] Z. He et al., "DADIGAN: A dual attention blocks-based disentangled iterative generative adversarial network for cloud and shadow removal on SAR and optical images," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103487.

[11] P. Wang, Z. He, B. Huang, M. Dalla Mura, H. Leung, and J. Chanussot, "VOGTNet: Variational optimization-guided two-stage network for multispectral and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9268–9282, May 2025.

[12] H. Chang, H. Bi, C. Xu, and F. Li, "LSFMamba: Local-enhanced spiral fusion mamba for multi-modal land cover classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 5, pp. 6081–6094, May 2026.

[13] D. Xu, W. Dong, S. Xiao, J. Qu, and Y. Li, "Diffusion feature completion-driven textual dual distillation for multimodal remote sensing classification with missing modality," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5515814.

[14] H. Chang, H. Bi, F. Li, C. Xu, J. Chanussot, and D. Hong, "Deep symmetric fusion transformer for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5644115.

[15] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509612.

[16] Y. Yang, J. Qu, W. Dong, L. Huang, and Y. Li, "Prototype-based meta-prompt tuning: Toward rehearsal-free few-shot class-incremental learning for multimodal remote sensing image," *IEEE Trans. Image Process.*, vol. 35, pp. 434–448, 2026.

[17] X. Cao, W. Dong, J. Qu, and Y. Li, "RMNet: Dual-dimensional difference recalibration-guided CNN-VMamba synergistic network for remote sensing image change captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5606913.

[18] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.

[19] W. Dong, Y. Yang, J. Qu, S. Xiao, and Y. Li, "Local information-enhanced graph-transformer for hyperspectral image change detection with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509814.

[20] S. Feng, S. Wang, C. Xu, C. Zhao, W. Li, and R. Tao, "Fractional domain information enhanced hyperspherical prototype learning method for hyperspectral image open-set classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5515117.

[21] W. Zhu, C. Zhao, S. Feng, and B. Qin, "Multiscale short and long range graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535815.

[22] R. Kumar, S. K. Roy, G. Vivone, S. Das, B. Raman, and P. Singh, "From classical image fusion to deep representation learning: A survey of the advances in hyperspectral image pan-sharpening," *Inf. Fusion*, vol. 127, Mar. 2026, Art. no. 103834.

[23] M. Ciotola et al., "Hyperspectral pansharpening: Critical review, tools, and future perspectives," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 1, pp. 311–338, Mar. 2025.

[24] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[25] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

[26] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[27] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS +pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[28] S. G. Mallat, "A theory for multiresolution signal decomposition: The wavelet representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 7, pp. 674–693, Jul. 1989.

[29] P. Wang et al., "Low-rank tensor completion pansharpening based on haze correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5405720.

[30] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[31] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[32] R. Dian, S. Li, L. Fang, T. Lu, and J. M. Bioucas-Dias, "Nonlocal sparse tensor factorization for semiblind hyperspectral and multispectral image fusion," *IEEE Trans. Cybern.*, vol. 50, no. 10, pp. 4469–4480, Oct. 2020.

[33] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, Jul. 2016. [Online]. Available: https://www.mdpi.com/2072-4292/8/7/594

[34] W. G. C. Bandara, J. M. J. Valanarasu, and V. M. Patel, "Hyperspectral pansharpening based on improved deep image prior and residual reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520816.

[35] Z. Jin, T. Zhang, T. Jiang, G. Vivone, and L. Deng, "LAGConv: Local-context adaptive convolution kernels with global harmonic bias for pansharpening," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2022, pp. 1113–1121.

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[36] J. He, J. Qu, X. Zhang, W. Dong, and Y. Li, "Diffusion-driven mutual enhancement of matching and fusion for reference-based hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5502813.

[37] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[38] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[39] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 103031–103063.

[40] J. Qu, J. Zhao, W. Dong, L. Zhang, and Y. Li, "MambaFedCD: Spatial–spectral–temporal collaborative mamba-based active federated hyperspectral change detection," *IEEE Trans. Image Process.*, vol. 35, pp. 1478–1492, 2026.

[41] T. Zhang, W. Dong, Y. Li, and J. Qu, "SAM-prior-guided spatial–spectral–temporal mamba framework for affine misregistration hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5506114.

[42] S. Peng, X. Zhu, H. Deng, L.-J. Deng, and Z. Lei, "FusionMamba: Efficient remote sensing image fusion with state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5410216.

[43] X. He et al., "Pan-mamba: Effective pan-sharpening with state space model," *Inf. Fusion*, vol. 115, Mar. 2025, Art. no. 102779. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253524005578

[44] X. Wang, Q. Hu, Y. Cheng, and J. Ma, "Hyperspectral image super-resolution meets deep learning: A survey and perspective," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 8, pp. 1668–1691, Aug. 2023.

[45] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and Z. Yu, "CNN-based hyperspectral pansharpening with arbitrary resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518821.

[46] L. He, J. Xie, J. Li, A. Plaza, J. Chanussot, and J. Zhu, "Variable subpixel convolution based arbitrary-resolution hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534519.

[47] J. Qu, Y. Chen, W. Dong, Y. Li, and Q. Du, "Adaptive coarse-to-fine parameter optimization for incremental hyperspectral target detection," *IEEE Trans. Circuits Syst. Video Technol.*, no. 99, Mar. 2026.

[48] X. Hu, H. Mu, X. Zhang, Z. Wang, T. Tan, and J. Sun, "META-SR: A magnification-arbitrary network for super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 1575–1584.

[49] Y. Chen, S. Liu, and X. Wang, "Learning continuous image representation with local implicit image function," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jul. 2021, pp. 8628–8638.

[50] J. Lee and K. H. Jin, "Local texture estimator for implicit representation function," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1919–1928.

[51] X. Xu, Z. Wang, and H. Shi, "UltraSR: Spatial encoding is a missing key for implicit image function-based arbitrary-scale super-resolution," 2021, *arXiv:2103.12716*.

[52] H.-W. Chen, Y.-S. Xu, M.-F. Hong, Y.-M. Tsai, H.-K. Kuo, and C.-Y. Lee, "Cascaded local implicit transformer for arbitrary-scale super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18257–18267.

[53] Y. Jiang et al., "HIIF: Hierarchical encoding based implicit image function for continuous super-resolution," 2024, *arXiv:2412.03748*.

[54] L. He, Z. Fang, J. Li, J. Chanussot, and A. Plaza, "Two spectral–spatial implicit neural representations for arbitrary-resolution hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5513621.

[55] G. Mai et al., "SSIF: Learning continuous image representation for spatial–spectral super-resolution," 2023, *arXiv:2310.00413*.

[56] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

[57] Y. Zheng, J. Li, Y. Li, J. Guo, X. Wu, and J. Chanussot, "Hyperspectral pansharpening using deep prior and dual attention residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8059–8076, Nov. 2020.

[58] S. Peng, C. Guo, X. Wu, and L.-J. Deng, "U2Net: A general framework with spatial–spectral-integrated double U-net for image fusion," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 3219–3227, doi: 10.1145/3581783.3612084.

[59] W. Dong, T. Zhang, J. Qu, S. Xiao, J. Liang, and Y. Li, "Laplacian pyramid dense network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507113.

[60] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1767–1777.

[61] J. Qu, X. Wu, W. Dong, J. Cui, and Y. Li, "IR&ArF: Toward deep interpretable arbitrary resolution fusion of unregistered hyperspectral and multispectral images," *IEEE Trans. Image Process.*, vol. 34, pp. 1934–1949, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10938029/

[62] L. He, Z. Fang, J. Li, H. Ye, and A. Plaza, "Arbitrary-resolution hyperspectral pansharpening neural operators," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510419.

[63] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," *Commun. ACM*, vol. 65, no. 1, pp. 99–106, Jan. 2022.

[64] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogrammetric Eng. remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[65] L. Alparone and A. Garzelli, "Benchmarking of multispectral pansharpening: Reproducibility, assessment, and meta-analysis," *J. Imag.*, vol. 11, no. 1, p. 1, Dec. 2024.

[66] G. Vivone et al., "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2015.

[67] L. Wald, *Data Fusion: Definitions and Architectures: Fusion of Images of Different Spatial Resolutions*. Paris, France: Presses des MINES, 2002.

[68] L. Alparone, B. Aiazzi, S. Baronti, A. Garzelli, F. Nencini, and M. Selva, "Multispectral and panchromatic data fusion assessment without reference," *Photogrammetric Eng. Remote Sens.*, vol. 74, no. 2, pp. 193–200, Feb. 2008.

[69] A. Garzelli and F. Nencini, "Hypercomplex quality assessment of multi/hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 662–665, Oct. 2009.

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